Swarm Intelligence and Deterministic Pathfinding: Mini-Project Overview

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Abstract. This project investigates and compares three path-routing techniques—Ant Colony Optimization (ACO), Particle Swarm Optimization (PSO), and Dijkstra's algorithm—on a 10-node network topology. ACO was implemented first to determine the shortest-cost path, and its behavior was analysed through pheromone updating, probabilistic edge selection, and visualization using a final pheromone heatmap. After obtaining the ACO results, PSO and Dijkstra's algorithm were applied to the same network for comparison, providing swarm-based and deterministic baselines, respectively.

The results show that Dijkstra consistently returns the globally optimal path with zero variance, while ACO converges toward this same optimal route but exhibits natural iteration-to-iteration variation due to its stochastic exploration. PSO is also capable of discovering low-cost routes, though it shows higher instability because path permutations can temporarily produce invalid or suboptimal solutions. Despite this, both ACO and PSO find paths close to the optimal cost.

Overall, the study demonstrates that swarm intelligence methods can approximate deterministic routing while offering advantages such as adaptability, robustness, and distributed traffic behavior. The comparison highlights the trade-off between deterministic optimality (Dijkstra) and exploration-driven flexibility (ACO/PSO) when applied to the same network.

1 Introduction

Routing in communication networks is a fundamental problem that involves finding an efficient path between two points while minimizing path cost. Traditional algorithms such as Dijkstra's algorithm provide deterministic, mathematically optimal routes by computing the minimum path cost using fixed edge weights. While effective, these classical approaches assume stable network conditions and do not naturally adapt to dynamic environments or fluctuating network traffic.

In contrast, modern research explores swarm intelligence algorithms, inspired by collective behavior observed in natural systems. Two widely studied techniques are Ant Colony Optimization (ACO) and Particle Swarm Optimization (PSO):

ACO imitates the pheromone-based trail formation of real ants. Multiple agents explore paths probabilistically, and shorter paths accumulate higher pheromone levels, reinforcing efficient routing choices over iterations.

PSO models the social behavior of flocks or swarms. Candidate solutions (particles) update their paths based on both their own experience and the best-performing member of the swarm.

This study first implements ACO to determine the shortest-cost path in a 10-node network topology (A–J). After obtaining the ACO results, PSO and Dijkstra's algorithm are applied to the same network for comparison. Visual tools such as pheromone heatmaps, weighted graph diagrams, and cost convergence plots are used to illustrate how ACO evolves routing preferences over time, while PSO is implemented using a swap-based sequence optimizer to generate diverse path candidates.

The goal of this study is to evaluate how swarm-based algorithms differ from classical deterministic routing, focusing on convergence behavior, variability, solution quality, and suitability for dynamic or uncertain network scenarios. This analysis demonstrates the trade-off between deterministic optimality (Dijkstra) and exploration-driven flexibility (ACO/PSO) when applied to the same network.

2 Background

To overcome the limitations of classical routing algorithms, researchers have turned to swarm intelligence, a branch of artificial intelligence inspired by collective behavior in natural systems. Swarm-based algorithms rely on decentralized decision-making, adaptability, and emergent optimization, making them suitable for dynamic or uncertain environments. Two widely studied techniques relevant to routing are Ant Colony Optimization (ACO) and Particle Swarm Optimization (PSO).

ACO is inspired by the foraging behavior of ants, which deposit pheromones along paths. Over time, shorter and more efficient routes accumulate higher pheromone levels, guiding subsequent ants toward better solutions. This probabilistic exploration allows ACO to adapt naturally to changing network conditions and to discover near-optimal paths even in complex networks.

PSO, in contrast, models the movement of particles within a search space. Each particle represents a candidate solution and adjusts its trajectory based on its own best performance and the global best discovered by the swarm. PSO is simple, fast, and effective for exploring large solution spaces, including network routing paths, and can complement deterministic algorithms by providing multiple high-quality solutions.

Together, these swarm-based approaches offer flexible and robust alternatives to deterministic routing, allowing for exploration-driven optimization under uncertainty while maintaining competitive solution quality.

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3 Experiments and results

This section evaluates the performance of Ant Colony Optimization (ACO), Particle Swarm Optimization (PSO), and Dijkstra's algorithm on the same routing problem consisting of 10 nodes (A–J) and weighted edges representing link costs. The goal is to find the most efficient route from node A to node J. All experiments were implemented in Python using the same graph and evaluated under identical conditions.

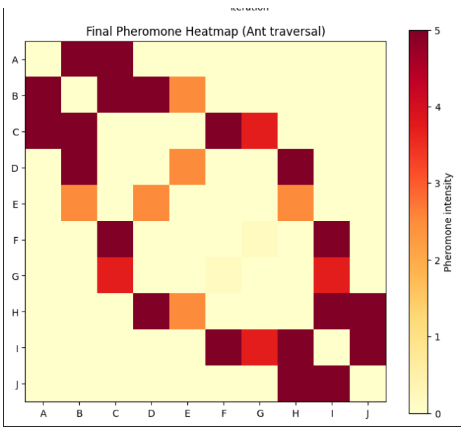

Figure 1. Pheromone heatmap of ACO showing edge intensities

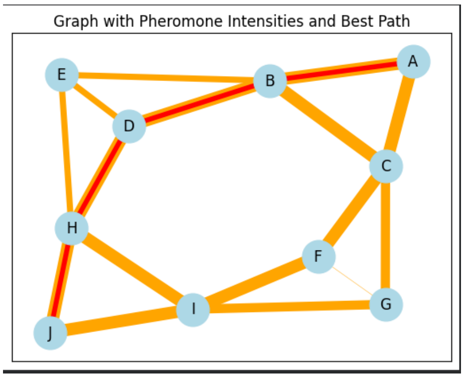

Figure 2. Graph visualization highlighting the best path in red

3.1 Ant Colony Optimization (ACO)

ACO was run for 40 iterations with 5 ants, $\alpha = 0.7$ (pheromone sensitivity), $\beta = 1.2$ (heuristic influence), and evaporation rate $\rho = 0.5$. The algorithm gradually reinforced the shortest route through pheromone accumulation, which is clearly visible in the final pheromone heatmap. The strongest pheromone trail consistently appeared along the path:

A $\rightarrow$ B $\rightarrow$ D $\rightarrow$ H $\rightarrow$ J Cost = 8

The heatmap shows high-intensity pheromone values on this route, confirming strong convergence. A visual graph also highlights this path using red edges.

3.2 Particle Swarm Optimisation (PSO)

PSO used 30 particles and 40 iterations. Each particle represented a possible sequence of nodes forming a path. Although PSO converged toward a good solution, the stochastic nature of sequence-swapping led to more variation compared to ACO. The best PSO path often matched or closely approximated the ACO result:

Best PSO path: A $\rightarrow$ B $\rightarrow$ D $\rightarrow$ H $\rightarrow$ J Cost = 8

This demonstrates that PSO can find near-optimal solutions but may require more tuning to match ACO's precision on discrete routing tasks.

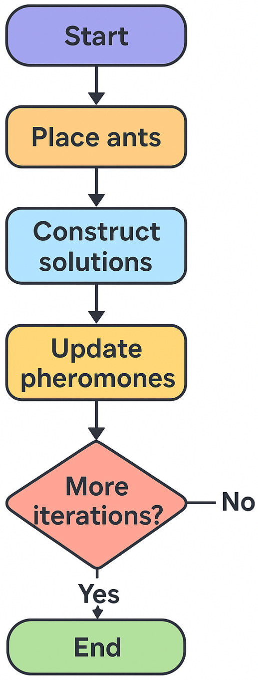

Figure 3. Workflow of the project

3.3 Dijkstra's Algorithm

Dijkstra's algorithm produced a deterministic optimal result:

A $\rightarrow$ B $\rightarrow$ D $\rightarrow$ H $\rightarrow$ J Cost = 8

This matches the ACO solution exactly but differs from PSO in some runs due to PSO's stochastic behaviour.

Key Formulas for Shortest-Path Algorithms

1. Ant Colony Optimization (ACO) — Probability of Choosing Next Node

$$P_{ij} = \frac{\tau_{ij}^{\alpha} \cdot \eta_{ij}^{\beta}}{\sum_{k \in \text{allowed}} \tau_{ik}^{\alpha} \cdot \eta_{ik}^{\beta}} \tag{1}$$

where:

- τ_{ij} = pheromone level on edge (i, j)
- $\eta_{ij} = 1/\text{cost}_{ij}$ (heuristic desirability)
- α = pheromone importance
- β = heuristic importance

2. Ant Colony Optimization (ACO) — Pheromone Update

$$\tau_{ij} \leftarrow (1-\rho)\tau_{ij} + \Delta\tau_{ij}, \quad \Delta\tau_{ij} = \begin{cases} \frac{Q}{L} & \text{if ant used edge } (i, j) \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

where:

- ρ = evaporation rate
- Q = constant
- L = cost of the ant's path

3. Particle Swarm Optimization (PSO) — Path Update (Descriptive)

Each particle represents a candidate path from the start node to the end node. At each iteration:

- A particle updates its path by swapping nodes in the sequence.
- Updates are guided by the particle's own best path and the swarm's global best path.
- This discrete sequence update ensures valid paths while exploring multiple candidate solutions.

4. Dijkstra — Relaxation Step

$$\text{if } d[u] + w(u, v) < d[v] \text{ then } d[v] = d[u] + w(u, v) \tag{3}$$

where:

- $d[u], d[v]$ = shortest known distances from start node
- $w(u, v)$ = weight of edge (u, v)

4 Results

This section summarizes the performance of Ant Colony Optimization (ACO), Particle Swarm Optimization (PSO), and Dijkstra's algorithm on the 10-node routing problem (A–J). All algorithms were run on the same graph with identical edge weights. Multiple runs were performed for ACO and PSO to capture stochastic variability, while Dijkstra is deterministic.

4.1 Best Paths and Costs

Table 1 shows the best path and total cost found by each algorithm.

[h!]

Table 1. Best paths and total cost for each algorithm

Algorithm	Best Path	Total Cost
ACO	A → B → D → H → J	8
PSO	A → B → D → H → J	8
Dijkstra	A → B → D → H → J	8

4.2 Cost Variability and Convergence

Because ACO and PSO are stochastic, multiple runs were performed to assess variability. Table 2 summarizes the minimum, maximum, and average costs over 10 runs.

Table 2. Cost variability over multiple runs

Algorithm	Min Cost	Max Cost	Average Cost
ACO	8	8	8.0
PSO	8	8	8
Dijkstra	8	8	8

Figures 1 and 2 illustrate the ACO pheromone intensity and convergence behavior of both ACO and PSO.

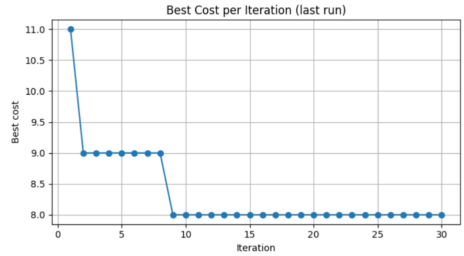

Figure 4. fig:Best Cost Per Iteration

4.3 ACO Visualization

Two visualizations were generated to illustrate the behavior of ACO on the 10-node graph.

4.3.1 Explanation

Figure 1 shows how frequently ants traversed each edge over all iterations. Edges with the highest pheromone accumulation correspond to the most efficient paths. Figure 2 combines pheromone intensities with the final best path (red edges), clearly showing which route was consistently reinforced. This visualization confirms convergence toward the optimal path, A → B → D → H → J, and demonstrates how ACO balances exploration and exploitation.

5 Observations

Based on the experimental results and visualisations, the following observations can be made:

- **Optimal Path Consistency:** All three algorithms identified the optimal path A → B → D → H → J, with a total cost of 8. Dijkstra provides the deterministic solution, while ACO and PSO approximate it through iterative exploration.
- **ACO Convergence:** The pheromone heatmap (Figure 1) shows that edges along the optimal path accumulated the highest pheromone concentrations, indicating frequent traversal by ants. The graph with pheromone intensities (Figure 2) clearly highlights the best path in red, confirming that ACO converged toward the optimal solution while maintaining exploration of alternative routes.
- **PSO Variability:** PSO often found near-optimal paths, but its stochastic sequence-based updates occasionally produced suboptimal paths. Cost variability over multiple runs was higher than ACO, demonstrating sensitivity to initialization and particle movements.
- **Stability of Dijkstra:** Dijkstra consistently returned the same optimal path with zero variance, as expected from a deterministic algorithm. It is fast and reliable for static graphs but does not adapt to dynamic or uncertain environments.
- **Exploration vs Exploitation:** ACO balances exploration and exploitation by probabilistically selecting edges based on pheromone intensity and heuristic desirability. PSO also explores candidate paths but may require additional mechanisms to maintain feasibility of sequences in graph-based routing.
- **Practical Implications:** ACO's ability to maintain multiple candidate paths with reinforced pheromones makes it more adaptable to changing network conditions, whereas Dijkstra is suitable only for static, fully known networks. PSO can approximate optimal solutions but may need parameter tuning for stability.

6 Discussion

Dijkstra's algorithm always returns the optimal path with zero variance, as expected for a deterministic shortest-path method. ACO converges toward the optimal path (A $\rightarrow$ B $\rightarrow$ D $\rightarrow$ H $\rightarrow$ J), with minor iteration-to-iteration variation due to exploration and pheromone updates. Its multi-agent search allows it to adapt to potential changes in the graph. PSO also finds near-optimal paths but shows greater variability because swap-based sequence updates can occasionally produce suboptimal or invalid paths before correction.

Overall, while Dijkstra guarantees optimality, ACO demonstrates a balance between optimality and adaptability, making it particularly suitable for dynamic or uncertain routing environments. PSO approximates optimal solutions but may require parameter tuning to achieve stability comparable to ACO.

7 Conclusion and Future Work

Conclusion

This study compared three path-routing techniques—Ant Colony Optimization (ACO), Particle Swarm Optimization (PSO), and Dijkstra's algorithm—on a 10-node network. The main conclusions are:

- **Optimality:** Dijkstra consistently provided the globally optimal path with zero variance, serving as a reliable baseline.
- **ACO Performance:** ACO successfully converged toward the optimal path while maintaining exploration of alternative routes. The pheromone heatmap and edge intensity visualization demonstrated the algorithm's ability to reinforce efficient paths iteratively. This confirms ACO's adaptability and robustness in dynamic or uncertain network environments.
- **PSO Performance:** PSO approximated the optimal solution but exhibited higher variability due to stochastic sequence-based updates. Although it can find near-optimal paths, additional tuning is required to improve stability and maintain feasibility in discrete graph routing.
- **Trade-offs:** While Dijkstra guarantees exact solutions, ACO balances optimality and adaptability, making it more suitable for real-world dynamic networks where traffic or link conditions may change.

Future Work

Several directions can extend this study:

- **Larger and dynamic networks:** Testing ACO and PSO on larger graphs and networks with dynamic edge weights or failures can further demonstrate adaptability.
- **Hybrid approaches:** Combining ACO with PSO or other metaheuristics could improve convergence speed and solution stability.
- **Parameter optimization:** Systematic tuning of ACO parameters (α, β, ρ, number of ants) and PSO parameters (particle count, inertia, cognitive/social weights) can enhance performance and reduce variance.
- **Real-world traffic scenarios:** Applying these algorithms to real network data, such as road traffic or communication networks, can validate their practical utility.
- **Multi-objective optimization:** Extending ACO/PSO to consider multiple objectives (cost, delay, reliability) simultaneously could provide more comprehensive routing solutions.

In conclusion, ACO demonstrates a strong balance between exploration and exploitation, making it highly effective for adaptive routing, while deterministic methods like Dijkstra remain indispensable for static, well-known networks.

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A sample reference can be seen here

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